21CSE306P · APPLIED GENERATIVE AI · COURSE PROJECT 2026–27

VERITAS.

Evidence-Driven Generative AI for Reliable Research

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Problem statement

Generative AI can produce **inaccurate or unsupported information** with high confidence. Existing AI research tools often lack **claim-level verification** and **conflict detection**, making it difficult to assess the reliability of generated information.

Retrieval-augmented generation grounds a model in documents, but the model can still add facts the evidence never states, and nobody checks each claim.

*VERITAS doesn't just answer: it **catches hallucinations claim by claim** and shows how much to trust every part of the answer.*

Objectives

- **01** Ground every statement in the user's own documents
- **02** Verify each claim individually against evidence
- **03** Detect conflicting evidence between sources
- **04** Give a reliable confidence score for every claim

End users

Students & researchers literature reviews with checked, citable answers

Teachers & reviewers see exactly where an AI answer comes from

Policy, health & legal teams catch unsupported statements in critical documents

Journalists & fact-checkers verify claims against the original sources

Data sources

User data PDF, DOCX, TXT or MD files uploaded by the user: the **only evidence** answers may use

Demo corpus 7 open-access arXiv papers on LLM hallucination (RAG, TruthfulQA, SelfCheckGPT, FActScore, …)

Benchmark **SciFact** (Wadden et al., 2020): 5,183 abstracts, 300 expert-labelled claims (340 claim–abstract pairs)

How VERITAS works: a fixed, single-pass generative pipeline

Knowledge indexing *once per document*	**Upload** The user's documents on any topic	→ **Clean & chunk** ~220-word passages with page numbers; bibliographies removed	→ **Embed** MiniLM turns each passage into 384 numbers that capture meaning	→ **Vector index** Passages stored for search by cosine similarity
Grounded generation *retrieval-augmented (RAG)*	**Question** Any research question about the selected documents	→ **Retrieve** Top-20 by meaning → cross-encoder re-rank → best 5 [S1–S5]	→ **Grounded prompt** Role + rules: use only the evidence, cite every sentence	→ **Generate** gpt-oss-120b writes the answer with [S#] citations
Verification & scoring *the core contribution*	**Claim extraction** gpt-oss-120b splits the answer into atomic, checkable facts	→ **NLI check** DeBERTa-v3: does the passage entail, contradict, or neither?	→ **Self-consistency** 4 re-sampled answers: is the fact repeated?	→ **Verified report** Verified · ! Uncertain · × Unsupported · ⊘ Contradicted

confidence = 0.6 × NLI + 0.4 × consistency − 0.5 × contradiction

Verified ≥ 0.70 · Unsupported < 0.40 · weak claims corrected **once** (no loops) · conflicting sources flagged

Concepts used

ML	Confidence fusion, thresholds, F1 / AUROC evaluation
DL	Neural embeddings for search by meaning
NLP	Chunking, claim splitting, inference (NLI)
Transformer	Bi-/cross-encoders, encoder & decoder
GenAI	RAG, sampling, prompt engineering

Models used

- GENERATOR · **gpt-oss-120b** (writes answers)
- VERIFIER · **DeBERTa-v3 NLI** (checks claims)
- EMBEDDER · **all-MiniLM-L6-v2** (search)
- RE-RANKER · **ms-marco-MiniLM-L-6** (ranking)

Prompt engineering

Role-play system persona	**Instructional** grounding rules
Few-shot completion claim extraction	**Classification** LLM-as-judge
Reasoning (CoT) verdict explanation	**Comparison** conflict explanation
Information closed-book baseline	**Creative** lay summary

Generative AI, not agentic

Agentic AI	VERITAS
plans its own steps · picks tools · loops until done	steps fixed in code · no tools · one single pass

Outputs & results

Generated output

"How can hallucinations in LLMs be detected?" (asked over 7 research papers)

Token-level classifiers trained on synthetic noisy data can label each token as hallucinated or not.	0.99
Hallucinatory outputs show concentrated attention and lower attention entropy.	0.96
"Generate-then-refine" pipelines can identify and correct hallucinations after generation.	0.92
Factored verification variants improve performance. *NLI 1.00, but only 31% of re-samples repeat it → confidence lowered*	0.72

Is the checker reliable?

DeBERTa-v3 vs. expert labels on SciFact (340 pairs)

0.70	0.69	0.86
accuracy (chance 0.33)	macro-F1	AUROC (chance 0.50)

	Supported	Contradicted	NEI
Supported	95	12	31
Contradicted	12	42	17
NEI	20	10	101

rows = experts · columns = model · NEI = not enough info

Prompt design vs. hallucination

Same questions, same evidence; only the prompt changed.

Bad prompt (vague)	74%
Zero-shot instructions	8%
Role + instructions (ours)	0%
Few-shot	17%
Chain-of-thought	0%

74% → 0% with prompt engineering alone (3-question preview)

Conclusion & future work

VERITAS turns a confident-but-unchecked LLM answer into a **claim-by-claim verified report**. Its checker agrees with experts well above chance, and prompt design alone removes most hallucination. **Next:** fine-tune the verifier on SciFact, learn the scoring weights, read tables and figures, and complete the full research study.

References

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